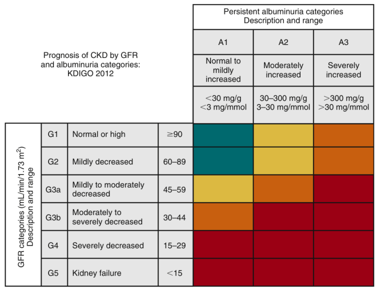


Green: low risk (if no other markers of kidney disease, no CKD)
Yellow: moderately increased risk
Orange: high risk
Red: very high risk

Fig. 54.1 Current classification system and nomenclature proposed by the Kidney Disease Improving Global Outcomes *(KDIGO)* in 2012. Chronic kidney disease *(CKD)* is defined as abnormalities of kidney structure or function, present for 3 months, with implications for health. CKD is classified on the basis of cause, glomerular filtration rate *(GFR)* category, and albuminuria category. (With permission from Kidney Disease: Improving Global Outcomes CKD Work Group. KDIGO 2012 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int*. 2013;3(Suppl):1–150.)

Fig_54_1

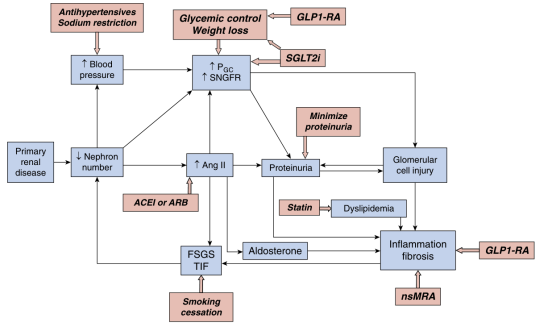


Fig. 54.2 A common pathway of mechanisms that result in a vicious circle of nephron loss in chronic kidney disease (CKD). Interventions (in *red*) for achieving kidney protection are directed at inhibiting the common pathway at multiple points to slow CKD progression. *ACEI,* Angiotensin-converting enzyme inhibitor; *Ang II,* angiotensin II; *ARB,* angiotensin receptor blocker; *FSGS,* focal segmental glomerulosclerosis; *GLP-1 RA,* glucagon-like peptide-1 receptor agonist; *nsMRA,* nonsteroidal mineralocorticoid antagonist; P_{GC}, glomerular capillary hydraulic pressure; *SGLT2i,* sodium-glucose contransporter 2 inhibitor; *SNGFR,* single nephron glomerular filtration rate; *TIF,* tubulointerstitial fibrosis.

Fig_54_2

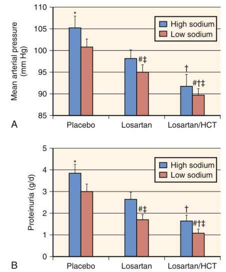


Fig. 54.3 Results of a prospective randomized crossover trial showing the effect of dietary sodium restriction on the (A) antihypertensive and (B) antiproteinuric effects of treatment with an angiotensin receptor blocker as monotherapy or in combination with a thiazide diuretic. *P < .05 versus all periods; #P < .05 versus same treatment on high-salt diet (effect of low-salt diet); †P < .05 versus losartan treatment on same diet [effect of hydrochlorothiazide *(HCT)*]; ‡P < .05 versus placebo on same diet. (From Vogt L, Waanders F, Boomsma F, et al. Effects of dietary sodium and hydrochlorothiazide on the antiproteinuric efficacy of losartan. *J Am Soc Nephrol*. 2008;19:999–1007.)

Fig_54_3

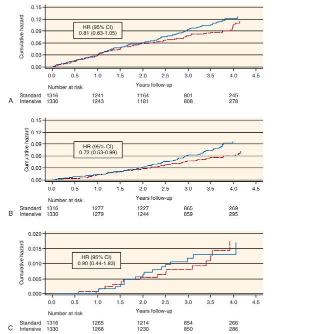
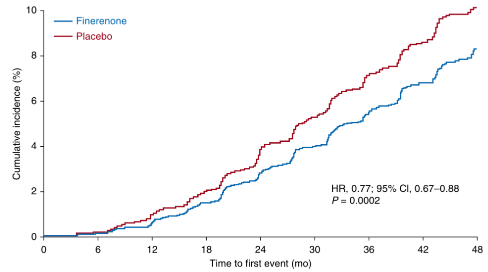


Fig. 54.4 Kaplan-Meier curves for prespecified outcomes in the Systolic Blood Pressure Intervention Trial (SPRINT) participants with chronic kidney disease. (A) The primary cardiovascular outcome, defined as the composite of myocardial infarction, acute coronary syndrome, stroke, acute decompensated heart failure, and death from cardiovascular causes. (B) The all-cause death outcome. (C) The main kidney outcome, defined as the composite of a decrease in estimated glomerular filtration rate of 50% or more from baseline (confirmed by repeat testing ≥90 days later) or the development of end-stage kidney disease. The *broken lines* depict the intensive blood pressure treatment group [systolic blood pressure (SBP) <120 mm Hg]; the *solid lines* depict the standard blood pressure treatment group (SBP <140 mm Hg). *CI,* Confidence interval; *HR,* hazard ratio. (From Cheung AK, Rahman M, Reboussin DM, et al. Effects of intensive BP control in CKD. *J Am Soc Nephrol*. 2017;28:2812–2823.)

Fig_54_4



Number of patients at risk

Finerenone	6519	6291	6107	5848	5027	3973	2815	2024	959
Placebo	6507	6292	6071	5815	4949	3932	2798	1988	962

Fig. 54.5 Results of a pooled analysis of the FIDELIO-DKD and FIGARO-DKD clinical trials of finerenone versus placebo in people with diabetic kidney disease. The analysis included 13,026 persons with a median follow-up of 3 years and showed that treatment with finerenone was associated with a 23% reduction in the cumulative incidence of the composite kidney outcome of time to kidney failure, sustained 57% or more decrease in eGFR from baseline, or kidney death. (From Bakris GL, Ruilope LM, Anker SD, et al. A prespecified exploratory analysis from FIDELITY examined finerenone use and kidney outcomes in patients with chronic kidney disease and type 2 diabetes. *Kidney Int*. 2023;103:196–206.)

Fig_54_5

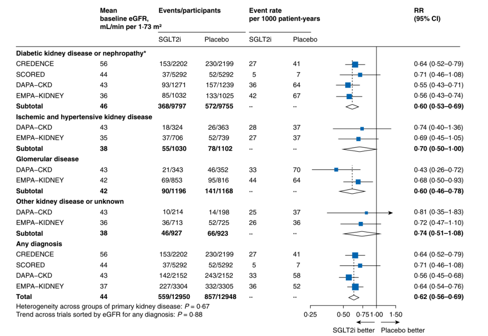


Fig. 54.6 Results of a meta-analysis of 13 randomized trials and 90,413 people of SGLT2 inhibitors versus placebo on the outcome of kidney disease progression. The analysis shows that the effect of SGLT2 inhibitors versus placebo is consistent, irrespective of the cause of primary kidney disease. (From Nuffield Department of Population Health Renal Studies Group; SGLT2 Inhibitor Meta-Analysis Cardio-Renal Trialists' Consortion. Impact of diabetes on the effects of sodium glucose co-transporter-2 inhibitors on kidney outcomes: collaborative meta-analysis of large placebo-controlled trials. *Lancet*. 2022;400(10365):1788–1801.)

Fig_54_6

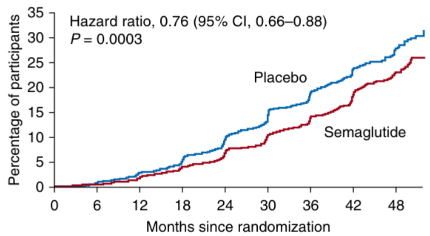


Fig. 54.7 Cumulative incidence graph demonstrating the effect of semaglutide versus placebo in 3533 people in the FLOW trial on the primary outcome of an eGFR decline ≥50%, kidney failure, or death from kidney disease or cardiovascular disease. (From Perkovic V, Tuttle KR, Rossing P, et al. Effects of semaglutide on chronic kidney disease in patients with type 2 diabetes. *N Engl J Med*. 2024;391(2):109–121.)

Fig_54_7

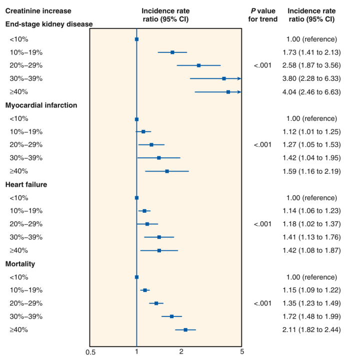


Fig. 54.8 Cardiorenal risks associated with levels of serum creatinine increase within 2 months after initiating treatment with renin–angiotensin–aldosterone system inhibitors. *CI,* Confidence interval. (From Schmidt M, Mansfield KE, Bhaskaran K, et al. Serum creatinine elevation after renin-angiotensin system blockade and long term cardiorenal risks: cohort study. *BMJ*. 2017;356:791.)

Fig_54_8

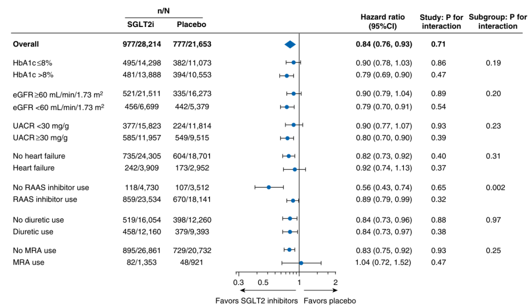


Fig. 54.9 Forest plot showing the results of a meta-analysis of 6 randomized trials involving 49,875 people of SGLT2 inhibitors versus placebo on the outcome of serious hyperkalemia, defined as a central laboratory determined serum potassium ≥6 mmol/L. The effect of SGLT2 inhibitors versus placebo on serious hyperkalemia is consistent, irrespective of the baseline characteristics of participants. *eGFR,* Estimated glomerular filtration rate; *HbA1c,* glycohemoglobin; *MRA,* mineralocorticoid receptor antagonist; *RAAS,* reninangiotensin-aldosterone system; *SGLT2i,* sodium-glucose cotransporter 2 inhibitor; *UACR,* urinary albumin-to-creatinine ratio. (From Neuen BL, Oshima M, Agarwal R, et al. Sodium-glucose cotransporter 2 inhibitors and risk of hyperkalemia in people with type 2 diabetes: a meta-analysis of individual participant data from randomized, controlled trials. *Circulation.* 2022;145(19):1460–1470.)

Fig_54_9

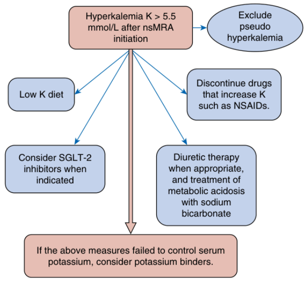


Fig. 54.10 A treatment algorithm for management of nsMRA related hyperkalemia. *NSAID,* Nonsteroidal anti-inflammatory drug; *nsMRA,* nonsteroidal mineralocorticoid receptor antagonist; *SGLT-2,* sodium-glucose cotransporter 2. (From Albakr RB, Sridhar VS, Cherney DZ. Novel therapies in diabetic kidney disease and risk of hyperkalemia: a review of the evidence from clinical trials. *Am J Kidney Dis*. 2023;82(6):737–742.)

Fig_54_10

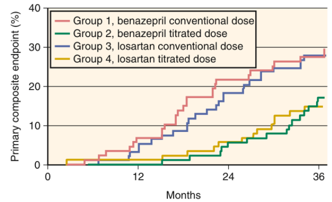


Fig. 54.11 Kaplan-Meier curves showing the incidence of the combined endpoint of doubling of serum creatinine level or end-stage kidney disease in persons with nondiabetic chronic kidney disease and urine protein levels exceeding 1 g/day. Persons in group 1 were randomly assigned to receive standard doses of angiotensin-converting enzyme (ACE) inhibitors; those in group 3 received standard doses of angiotensin receptor blockers (ARBs); those in group 2 received ACE inhibitors titrated upward to the maximum antiproteinuric levels; and those in group 4 received ARBs titrated upward similarly. (From Gruppo Italiano di Studi Epidemiologici in Nefrologia [GISEN]. Randomised placebo-controlled trial of effect of ramipril on decline in glomerular filtration rate and risk of terminal renal failure in proteinuric, non-diabetic nephropathy. *Lancet*. 1997;349:1857–1863.)

Fig_54_11

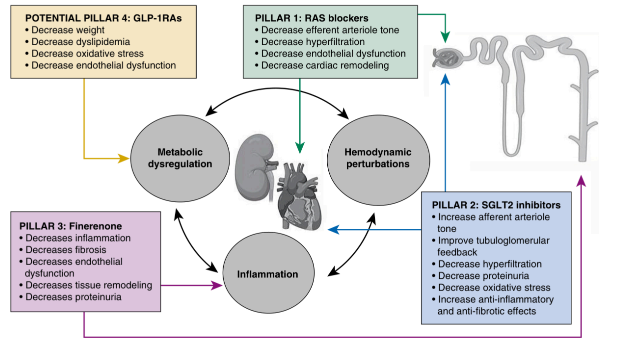


Fig. 54.12 The pillars of treatment for diabetic kidney disease to slow the progression of chronic kidney disease. GLP-1 RA, glucagon-like peptide-1 receptor agonist; RAS, renin-angiotensin system; SGLT-2, sodium-glucose cotransporter 2. (From Naaman, S. C. and G. L. Bakris."Diabetic Nephropathy: Update on Pillars of Therapy Slowing Progression." Diabetes Care 2023 46(9): 1574-1586.)

Fig_54_12

Table 54.1 Summary of Studies Showing the Kidney-Protective Effects of RAAS Inhibitors, SGLT2 Inhibitors, nsMRAs, and GLP1RAs in Diabetic and Nondiabetic Persons With Chronic Kidney Disease

CKD Type	Trial Outcome	Reference
Angiotensin-Converting Enzyme Inhibitors		
Type 1 DM + CKD	↓ Risk of dialysis or death	Lewis et al.[114]
Type 1 DM + microalbuminuria	↓ Risk of overt nephropathy	Mathiesen et al.[301] Laffel et al.[302] Viberti et al.[303]
Type 1 DM + normoalbuminuria	No significant benefit	Mauer et al.[130] EUCLID Study Group[304]
Type 2 DM + CKD	Benefit in 1 study only	Bakris et al.[305]
Type 2 DM + microalbuminuria	↓ Risk of overt nephropathy	Ravid et al.[119,120] Sano et al.[116] Trevisan and Tiengo[116] Agardh et al.[118] Ahmad et al.[121] Ruggenenti[128]
Type 2 DM + normoalbuminuria	↓ Risk of developing microalbuminuria	Ravid et al.[122]
Nondiabetic CKD	↓ Doubling of creatinine level/ESKF	Ruggenenti et al.[138,139]
Angiotensin Receptor Blockers		
Type 1 DM + normoalbuminuria	Small ↑ in albuminuria or no benefit	Mauer et al.[130] Bilous et al.[133]
Type 2 DM + normoalbuminuria	↓ Risk of developing microalbuminuria (small ↑ in rate of cardiovascular mortality)	Bilous et al.[133] Haller et al.[134]
Type 2 DM + microalbuminuria	↓ Risk of overt nephropathy	Parving et al.[125]
Type 2 DM + CKD	↓ Risk of doubling of creatinine level ↓ Risk of ESKF	Lewis et al.[124] Brenner et al.[123]
Sodium-Glucose Cotransporter 2 Inhibitors		
Type 2 DM and nondiabetic CKD	↓ Risk of CKD progression, ESKF, or death from kidney failure	Perkovic et al.[170] Heerspink et al.[171] EMPA-KIDNEY[173]
Nonsteroidal Mineralocorticoid Receptor Antagonist		
Type 2 DM + albuminuria	↓ Risk of CKD progression, death from kidney failure, ESKF or transplantation	Bakris et al.[160]
Glucagon-Like Peptide-1 Receptor Agonist		
Type 2 DM + albuminuria	↓ Risk of CKD progression, death from kidney failure, ESKF or transplantation	Perkovic et al.[184]

CKD, Chronic kidney disease; *DM,* diabetes mellitus; *ESKF,* end-stage kidney disease; *GLP1RA*, glucagon-like peptide-1 receptor agonist;

Table_54_1

Table 54.2 Overview of Chronic Kidney Disease Management by Stage

Features	Stages 1 and 2	Stage 3a	Stage 3b	Stage 4	Stage 5
Estimated GFR	≥60 mL/min/1.73 m² + albuminuria or hematuria or structural kidney damage	45-59 mL/min/1.73 m²	30-44 mL/min/1.73 m²	15-29 mL/min/1.73 m²	<15 mL/min/1.73 m²
Frequency of monitoring	Annual	6-12 months	3-6 months	3-4 months	1-3 months
Laboratory testing	Annual electrolytes and estimated GFR Annual urine ACR (or other estimate of proteinuria) Baseline anemia and mineral and bone profiles Glucose, lipids, and HbA_{1c} See Table 54.3 for causes of AKI after initiation of ACE inhibitor or ARB therapy.				Check electrolytes and estimated GFR 1 week after new use or higher doses of ACE inhibitors or ARBs; otherwise, assess electrolytes/estimated GFR, mineral and bone, and anemia profiles every 3-6 months, depending on GFR decline.
Blood pressure targets	BP target <130/80 mm Hg with proteinuria BP target <140/90 mm Hg without proteinuria if no clinical or radiologic evidence of ARVD or previous episodes of AKI				Risk of AKI is increased in elderly persons (>75 years), those with CHF, and those with ARVD; 140/90 mm Hg may be more appropriate for these groups.
Blood pressure agents	ACE inhibitor or ARB if urine ACR ≥30 mg/g Most persons need 2-4 agents in total to achieve these targets, in a combination of ACE inhibitors or ARBs and 1 or more of the following: a diuretic (all classes), a calcium channel blocker, and a β-blocker.				Loop diuretics are now usually required for BP and edema control.
SGLT2 Inhibitors	SGLT2 inhibitors in all people with CKD and type 2 diabetes and eGFR ≥20 mL/min/1.73 m² SGLT2 inhibitors in all people with CKD without diabetes and an eGFR between 20 and 45 mL/min/1.73 m² irrespective of UACR or an eGFR of ≥45 mL/min/1.73 m² with a UACR ≥200 mg/g				Continue SGLT2 inhibitor.
nsMRA	nsMRA for all people with CKD with type 2 diabetes and an eGFR ≥25 mL/min/1.73 m², normal serum potassium concentration, and a UACR ≥30 mg/g despite maximum tolerated dose of ACE inhibitor or ARB				Continue nsMRA as tolerated.
Cardiovascular prevention	Statin if CVD risk ≥10% over 10 years HbA_{1c} <7.0% unless at risk for severe hypoglycemia GLP1RA as the preferred second-line glucose lowering medication for people with diabetes at high cardiovascular risk			Consider statin for all persons. GLP1RA for people with diabetes at high cardiovascular risk	Continue statin.
Bone and anemia complications	If PTH level rises progressively, commence phosphate restriction and then consider therapy with vitamin D or analog. If anemia is out of keeping with GFR, confirm or rule out gastrointestinal blood loss.				Give intravenous iron before ESA if hemoglobin count <10 g/dL. Maintain target hemoglobin count of 10-11.5 g/dL.
Lifestyle and nutritional management	Smoking cessation Moderate exercise up to 30-60 min/day 4-7 days/wk Target weight with BMI <25 kg/m² Reduced salt intake as per DASH diet <5 g/day GLP1RA as the preferred second line glucose lowering medication for people with diabetes and requiring weight management				Limit dietary potassium excess. Weigh at each clinic and assess fluid overload, anorexia, physical function.
Specific RRT planning steps	Education regarding progression and role of conservative management regarding blood pressure targets and specific primary kidney disease treatment if indicated Hepatitis B vaccination if risk of progression is high			Education on RRT types and palliative care if CKD is progressing or patient is at high risk of progression. Hepatitis B vaccination.	AVF creation PD catheter insertion Enter on list for transplant.

Continued on following page

Table_54_2_part1

Table 54.2 Overview of Chronic Kidney Disease Management by Stage (Continued)

Features	Stages 1 and 2	Stage 3a	Stage 3b	Stage 4	Stage 5
Referral guidance from primary care physician to nephrologist	Progressive or abrupt fall in estimated GFR Proteinuria (urine protein levels >0.5 g/day; ACR >300 mg/g or >30 mg/mmol)			Refer unless patient is terminally ill	Refer unless patient is terminally ill.

ACE, Angiotensin-converting enzyme; *ACR,* albumin-to-creatinine ratio; *AKI,* acute kidney injury; *ARB,* angiotensin receptor blocker; *ARVD,* atherosclerotic renovascular disease; *AVF,* arteriovenous fistula; *BMI,* body mass index; *BP,* blood pressure; *CHF,* congestive heart failure; *CKD,* chronic kidney disease; *CVD,* cardiovascular disease; *DASH,* Dietary Approaches to Stop Hypertension; *ESA,* erythropoietin-stimulating agent; *GFR,* glomerular filtration rate; *GLP1RA,* glucagon-like peptide 1 receptor agonists; *HbA_{1c}*, hemoglobin A_{1c}; *PD,* peritoneal dialysis; *nsMRA,* nonsteroidal Mineralocorticoid Receptor Antagonist; *PTH,* parathyroid hormone; *RAAS,* renin–angiotensin–aldosterone system; *RRT,* renal replacement therapy; *SGLT2,* sodium-glucose cotransporter-2.

Table_54_2_part2

Table 54.3 Causes of Acute Kidney Injury After Initiation of Therapy With Angiotensin-Converting Enzyme Inhibitor or Angiotensin Receptor Blocker

Blood Pressure Insufficient for Adequate Renal Perfusion

Poor cardiac output
Low systemic vascular resistance (e.g., as in sepsis)
Volume depletion (gastrointestinal loss, poor oral intake, excess diuretic use)

Presence of Renal Vascular Disease[a]

Bilateral renal artery stenosis
Stenosis of dominant or single kidney
Afferent arteriolar narrowing (caused by hypertension and cyclosporine)
Diffuse atherosclerosis in smaller renal vessels

Vasoconstrictor Agents (Nonsteroidal Anti-inflammatory Drugs, Cyclosporine)

[a]Clinical features of renal vascular disease include vascular bruits (areas of the epigastric, femoral, and carotid arteries), prior rise in serum creatinine level of more than 30%, fall in estimated glomerular filtration rate of more than 20% after beginning of treatment with an angiotensin-converting enzyme inhibitor or angiotensin receptor blocker, and a history of flash pulmonary edema.

Table_54_3

Table 54.4 Comprehensive Strategy and Therapeutic Goals for Achieving Maximal Kidney Protection in Persons With Chronic Kidney Disease

Intervention	Goals
ACE inhibitor or ARB treatment	UACR <30 mg/g (<3 mg/mmol) GFR decline <1 mL/min/year
Additional antihypertensive therapy	BP <130/80 mm Hg if urine albumin excretion >30 mg/day BP <140/90 mm Hg if urine albumin excretion <30 mg/day
Start SGLT2 inhibitors, nsMRA, and GLP1-RA in all eligible patients	UACR <30 mg/g (<3 mg/mmol) GFR decline <1 mL/min/year
Weight loss if patient is overweight	Aim for 5% weight loss
Dietary salt restriction	<5 g/day (equivalent to 90 mEq sodium/day)
Dietary protein restriction	Avoid high-protein intake >1.3 g/kg/day In stages 4 and 5 CKD, consider reducing intake to 0.8 g/kg/day
Tight glycemic control	HbA_{1c} <7.0% unless considered high risk for hypoglycemia
Smoking cessation	Complete cessation
Lipid-lowering therapy	Total cholesterol <200 mg/dL (5.2 mmol/L) LDL cholesterol <100 mg/dL (2.6 mmol/L)

ACE, Angiotensin-converting enzyme; *ARB,* angiotensin receptor blocker; *BP,* blood pressure; *CKD,* chronic kidney disease; *GFR,* glomerular filtration rate; *GLP1RA*, glucagon-like peptide 1 receptor agonists; *HbA_{1c}*, hemoglobin A_{1c}; *LDL,* low-density lipoprotein; *nsMRA*, nonsteroidal Mineralocorticoid Receptor Antagonist; *UACR*, urine albumin-to-creatinine ratio.

Table_54_4

Table 54.5 Recommended Frequency of Monitoring by Stage of Chronic Kidney Disease

Variable	Stage 1 and 2	Stage 3	Stage 4	Stage 5
GFR and electrolytes	Every 12 months	Every 3-12 months	Every 3-6 months	Every 1-3 months
Proteinuria with ACR or PCR testing	Every 12 months	Every 3-12 months	Every 3-6 months	Every 3-6 months
Blood pressure	Each visit	Each visit	Each visit	Each visit
Calcium and phosphate levels	Every 12 months	Every 12 months	Every 3-6 months	Every 3 months
Parathyroid hormone level	—	Every 12 months	Every 3-6 months	Every 3-6 months[a]
Hemoglobin	Every 12 months	Every 12 months	Every 3-6 months	Every 1-3 months[a]

[a]Monitoring of parathyroid hormone and anemia should depend on the previous results and specific treatment, if any, for those conditions. Stable values with no specific treatment require less monitoring as indicated.
Monitoring should be individualized according to previous rate of GFR decline, risk assessment of future GFR decline (particularly high if heavy proteinuria, >1 g/day or equivalent), and current drug therapy.
ACR, Albumin-to-creatinine ratio; *GFR,* glomerular filtration rate; *PCR,* protein-to-creatinine ratio.

Table_54_5